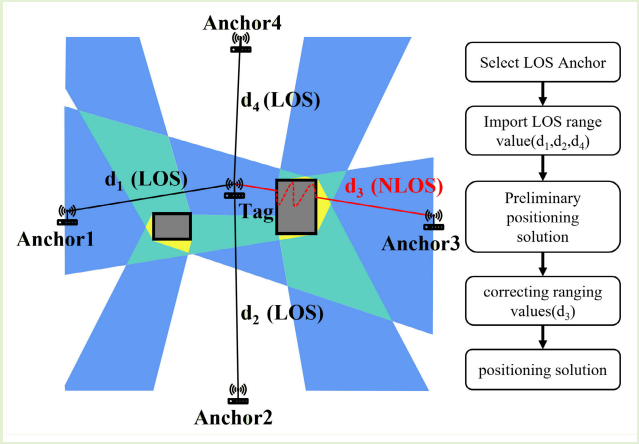
An Indoor UWB NLOS Correction Positioning Method Based on Anchor LOS/NLOS Map

Qing Wang, Zehui Li, Hao Zhang, Yuan Yang, and Xiaolin Meng

***Abstract*—With the increasing demand for location-based services (LBS) in daily production and life, indoor positioning has gradually become one of the current research hotspots. The most pressing challenge in indoor positioning is achieving high-precision location in complex indoor environments. Aiming at the problem that severe non-line-of-sight (NLOS) interference with ultrawideband (UWB) positioning accuracy in complex indoor environments, this article proposes a novel indoor UWB NLOS correction positioning method based on anchor LOS/NLOS map, which utilizes prior environmental information. After anchors are deployed, the LOS/NLOS map is generated using the prior environmental information. Then, the line-of-sight (LOS) anchors are selected for position solution, and the observations under the influence of the NLOS anchors are corrected. Then, the trajectory is optimized using a map-modified Kalman filtering algorithm, and it adopts a method to solve the position of two LOS anchors by using trajectory constraints, which can deal with more complex indoor environment. The experimental results show that the NLOS correction method in this article has better accuracy than the traditional filtering algorithm, especially in NLOS environment with few anchors. Compared with the standard Kalman filtering algorithm, the average positioning accuracy is improved by 69.30%, and compared with the robust Kalman filtering algorithm, the average positioning accuracy is improved by 33.77%.**



***Index Terms*— Indoor positioning, line-of-sight (LOS)/non-line-of-sight (NLOS) map, ultrawideband (UWB).**

I. Introduction

WITH the development of smart cities and the Internet of Things, intelligent devices and technologies are rising. The social and commercial value generated by location-based services (LBS) is increasing significantly, which has attracted more and more attention. Global navigation satellite system (GNSS) has been used for LBS for a long time. It provides accurate and reliable location information in outdoor environment and has a very high utilization rate in outdoor positioning. In the indoor environment, it is difficult to receive GNSS signals due to building occlusion. Therefore, NLOS is serious in indoor environment and is the main factor affecting positioning accuracy [1].

At present, among the indoor mainstream positioning technologies, there are received signal strength indicators (RSSIs) [2], [3], time of flight (TOF) [4], angle of arrival (AOA) [5], and channel status information (CSI) [6], [7], [8]. Among them, RSSI fingerprint information positioning methods, represented by Wi-Fi and Bluetooth, are extremely vulnerable to environmental impact [9], [10], [11] and need to update the fingerprint database regularly, which is difficult to maintain, and the location accuracy is not high. The positioning mode based on AOA and CSI requires the support of MIMO array antenna, and the hardware complexity is higher. With the development of UWB technology, its positioning accuracy can reach the centimeter level, so the UWB indoor positioning technology using the TOF method on high-precision indoor positioning system is more popular with the public [12], [13].

Existing literature indicates that in open indoor environments, using UWB signals for line-of-sight (LOS) transmission can achieve centimeter-level positioning accuracy, which fully meets people's demand for LBS indoors [14]. In complex

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indoor environments, motion tags may not receive the LOS signals of UWB but only receive multipath signals, such as reflections and diffractions, where the UWB signals propagate in non-line-of-sight (NLOS) conditions [15]. In the NLOS area, it is obvious that the signal propagation time will be extended regardless of reflection or diffraction propagation, resulting in a large NLOS error in the calculation of ranging data [16]. NLOS error is unavoidable in daily indoor environment, and the Kalman filter with NLOS mitigation has been studied by many researchers [17], [18], [19]. In [17], unbiased and biased Kalman filters were utilized to estimate time of arrival (TOA) based on NLOS identification results. The smoothed TOA estimates were then converted to time-difference-of-arrival (TDOA) estimates for determining the position. Liao and Chen [18] proposed an interactive multiple model smoother based on Kalman filter for TOA data smoothing. A bimodal Markov process was used to describe the transitions between LOS and NLOS propagation conditions for the moving tag and each anchor. Yu and Dutkiewicz [19] presented a distance smoothing method utilizing online LOS and NLOS identification and using mean and STD estimates to update the Kalman filter equation for distance measurement smoothing. Due to the presence of NLOS range measurements, it is not ideal to use the traditional Kalman filter algorithm to solve the NLOS positioning error. Therefore, how to weaken the NLOS error or correct the NLOS ranging value has become the focus of many scholars' research [20], [21]. Scholars have mainly adopted the following solutions to optimize the NLOS positioning: reducing the impact of NLOS observations by using more anchors [22]. Adding robustness factor to optimize the Kalman filter algorithm to complete the identification of NLOS signals and the positioning solution is computed after discarding NLOS observations [23], [24], [25]. Modeling the NLOS error environment is to correct NLOS observations [26], [27]; using multisensor fusion, especially IMU and UWB fusion positioning, is to reduce the impact of NLOS error on UWB positioning [15], [28], [29], [30], [31]; and neural network learning is used to distinguish and mitigate NLOS errors [13], [32], [33], [34]. However, among the above methods, increasing LOS anchors and joining the IMU fusion positioning methods obviously increase the positioning cost, and the algorithm needs to process more raw data, which increases the time complexity; an appropriate robust factor needs to be set in the robust algorithm to have a good robust effect; otherwise, the results will diverge. Moreover, the method of discarding NLOS observations reduces the necessary input observations, which produces prediction bias; in theory, the method of establishing the environment model of NLOS error can correct the NLOS observations to achieve the positioning effect in the LOS environment. However, most of the current NLOS error environment models are empirical models, which are susceptible to environmental impact, and a deterministic model with universality is not produced; the learning methods of neural network need to train a large number of datasets, and most of them combine with fingerprint positioning methods, so the computation is heavy and the real-time positioning cannot be guaranteed.

Based on the above problems, we propose a novel indoor UWB NLOS correction positioning method based on anchor LOS/NLOS map, which utilizes prior environmental information. In the NLOS area, the LOS anchors are used to complete the preliminary positioning solution, and the observations of the NLOS anchors are corrected to optimize the final positioning solution. At the same time, to our knowledge, the current research has not proposed an effective solution for the positioning of few anchors. In this article, when there are fewer than three LOS anchors for traditional positioning, a position-constrained two anchors auxiliary positioning algorithm is used to correct NLOS observations.

The main contributions of this article are given as follows.

1) The map information is introduced into the indoor positioning, and the anchor LOS/NLOS map is constructed. LOS anchors are selected by the map to complete the preliminary positioning solution, and then, the ranging values of NLOS anchors are corrected.
2) In the case of fewer LOS anchors, a position-constrained two anchors auxiliary positioning algorithm is adopted to complete the correction of NLOS observations.
3) Based on the anchor LOS/NLOS map, combined with the Kalman filter algorithm, a high-precision UWB positioning method for handling NLOS error in indoor complex environment is constructed. The performance of the system is verified through simulation test and actual experiment.

The remainder of this article is organized as follows. In Section II, we propose a map-constrained multianchor positioning optimization method and a few-anchor positioning optimization method. In Section III, two traditional trajectory optimization algorithms are described, and a map-constrained multianchor trajectory optimization algorithm is presented. Section IV presents the simulation experiment and experimental analysis. Section V presents the actual experiment and experimental analysis. Finally, Section VI concludes this article.

II. Multianchor Positioning Optimization Algorithm With Map Constraints

In complex indoor environments, such as underground garage, large shopping malls, and office buildings, the NLOS environmental impact formed by the building structure is the main factor that causes the positioning accuracy of complex indoor environment to be less than that of open environment. It is difficult for indoor positioning researchers to solve this problem, but GIS workers can build semantic maps with rich environmental information [35]. Therefore, we can consider extracting useful information from semantic maps for complex indoor positioning. We noticed that once our anchors deployment is complete, we can quickly get the LOS and NLOS regions of each anchor across the map. From this, we can build an LOS/NLOS map of UWB anchor to distinguish the tag corresponding to LOS anchor and NLOS anchor anywhere in the map area, and on this basis, the NLOS ranging values can be corrected.

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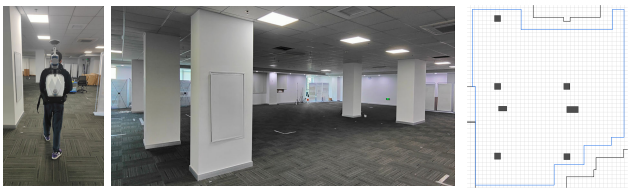


Fig. 1. Mapping experimental scenes: (a) LiDAR backpack, (b) experimental site, and (c) floor plan.

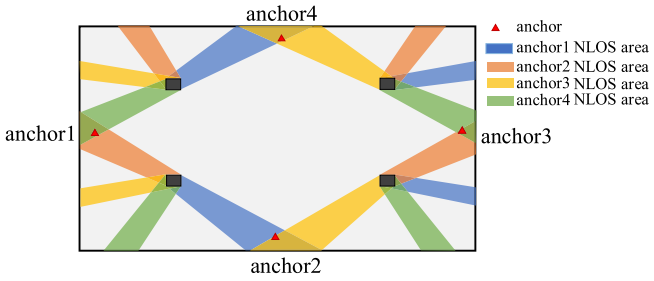


Fig. 2. Anchor LOS/NLOS map.

A. UWB Anchor LOS/NLOS Map Acquisition

With the development and application of lidar equipment, it is no longer difficult to collect and map indoor spatial structure information. Using a lidar backpack device that combines two 16-line velodyne lidar and panoramic cameras as shown in Fig. 1(a), this article collects information from Room 309 with NLOS characteristics similar to those of typical spatial structures of metro stations and underground parking lots, as shown in Fig. 1(b). The collection process takes about 2 min to generate the laser point cloud data. Using the modeling software, a simple 2-D plan of the experimental site can be quickly generated, and the spatial dimensions of the site can be obtained, as shown in Fig. 1(c). Seven gray rectangles in the plot are the columns in the site, with a square column edge of 0.8 m. The seven gray rectangles in the figure are the columns in the site, of which the side length of the square column is 0.8 m.

Once site information is obtained, it is determined whether the segment from the anchor to the current location intersects the obstacle. From this, it can be determined whether the anchor is an LOS anchor for the current location. As shown in Fig. 2, four anchors are located at points A, B, C, and D, four columns in fixed space act as obstacles, and the NLOS areas are shown in shaded areas. When the tag moves to the corresponding NLOS area, if the NLOS ranging value is discarded, the influence of the NLOS error on the positioning result can be eliminated from the source.

B. UWB Positioning Solution Method

When there are three or more LOS anchors, the least squares solution of the polygonal ranging algorithm can be used as the positioning coordinates for the solution. The schematic is shown in Fig. 3.

In the multilateral ranging shown in the figure above, the geometric relationship between the anchors and the tag

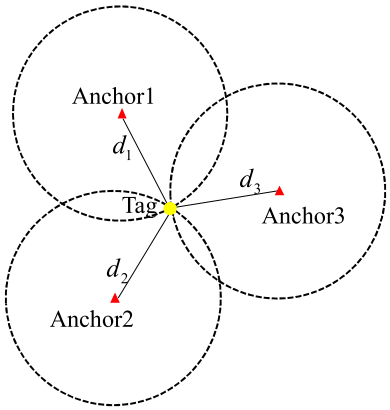


Fig. 3. Principle of multilateral positioning method.

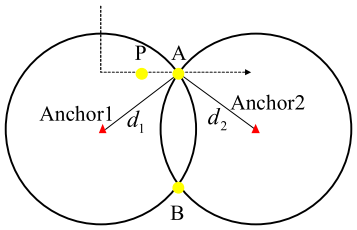


Fig. 4. Two anchor positioning solution.

satisfies the following equation:

$$\begin{aligned} d_1^2 &= (x - x_1)^2 + (y - y_1)^2 \\ d_2^2 &= (x - x_2)^2 + (y - y_2)^2 \\ d_3^2 &= (x - x_2)^2 + (y - y_3)^2 \end{aligned} \tag{1}$$

where d_i is the distance from the tag to the i anchor, (x, y) is the motion tag, and (x_i, y_i) is the corresponding anchor coordinate.

Organize the above formula and write it in the form of matrix, as shown in the following equation:

$$2 \cdot \begin{bmatrix} x_1 - x_2 & y_1 - y_2 \\ x_1 - x_3 & y_1 - y_3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} d_2^2 - d_1^2 + x_1^2 + y_1^2 - x_2^2 - y_2^2 \\ d_3^2 - d_1^2 + x_1^2 + y_1^2 - x_3^2 - y_3^2 \end{bmatrix}. \tag{2}$$

Since the ranging value $\hat{d}$ is used in actual calculation, the error of ranging value can be reduced by obtaining the least square solution of (2), as shown in the following equation:

$$X = \left(A^T A\right)^{-1} A^T \eta \tag{3}$$

where

$$X = \begin{bmatrix} x \\ y \end{bmatrix}, \quad A = 2 \cdot \begin{bmatrix} x_1 - x_2 & y_1 - y_2 \\ x_1 - x_3 & y_1 - y_3 \end{bmatrix}$$

$$\eta = \begin{bmatrix} d_2^2 - d_1^2 + x_1^2 + y_1^2 - x_2^2 - y_2^2 \\ d_3^2 - d_1^2 + x_1^2 + y_1^2 - x_3^2 - y_3^2 \end{bmatrix}.$$

When there are fewer than three LOS anchors, the traditional polygonal ranging algorithm is invalid in the 2-D plane. For this reason, this article proposes a position-constrained two anchors auxiliary positioning algorithm, as shown in Fig. 4.

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Obviously, a circle formed by the ranging values of the two anchors in a 2-D plane produces two intersection points, namely, A and B, as shown in the figure, whereas a tag position can only be A or B, so it is not reasonable to use least squares at this time. Solution positions can be constrained by using the position of the previous moment in the tag motion track. Assuming that the position at the last moment is $P(x_{t-1}, y_{t-1})$, that is, a point with a smaller distance between PA and PB is selected as the current position solution point and the equation is given as follows:

$$\begin{aligned} d_1^2 &= (x - x_1)^2 + (y - y_1)^2 \\ d_2^2 &= (x - x_2)^2 + (y - y_2)^2 \\ d_{t-1,t}^2(x, y) &= (x - x_{t-1})^2 + (y - y_{t-1})^2 \end{aligned} \tag{4}$$

where $d_{t-1,t}$ is the distance between the current position and the position at the previous time and the smaller (x, y) is taken as the solution coordinate, namely, $\arg\min\{d_{t-1,t}^2(x, y)\}$.

III. Multianchor Trajectory Optimization Algorithm With Map Constraints

The previous section mainly introduced the positioning solution method for quickly determining LOS/NLOS anchor and how to solve the positioning problem with few LOS anchors. The following describes the positioning trajectory optimization algorithm of tag in map constraint.

A. Standard Kalman Filter

The standard Kalman filter (SKF) algorithm is the most basic trajectory constraint algorithm, which can smooth the trajectory [17]. This article takes four-anchor positioning as an example to introduce a simple process.

The state vector is the motion state of tag, as shown in the following equation:

$$X_k = \begin{bmatrix} x_k & \dot{x}_k & y_k & \dot{y}_k \end{bmatrix}^T \tag{5}$$

where x_k and y_k are the coordinate positions of tag and $\dot{x}_k$ and $\dot{y}_k$ are the velocities in the x- and y-directions, respectively. The state equation of the algorithm is given as follows:

$$X_{k+1} = AX_k + \omega_k \tag{6}$$

$$A = \begin{bmatrix} 1 & T & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & T \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$Q = \begin{bmatrix} \frac{T^4}{4} & \frac{T^3}{2} & 0 & 0 \\ \frac{T^3}{2} & T^2 & 0 & 0 \\ 0 & 0 & \frac{T^4}{4} & \frac{T^3}{2} \\ 0 & 0 & \frac{T^3}{2} & T^2 \end{bmatrix} \sigma^2 \tag{7}$$

where A is the state transition matrix, ω_k is the process noise at time k, T is the sampling time, Q is the process noise covariance, and σ is the process noise power spectral density.

According to the equation in (2), the observation equation of the algorithm is shown in the following equation:

$$\begin{aligned} 2 \cdot \begin{bmatrix} x_1 - x_2 & 0 & y_1 - y_2 & 0 \\ x_1 - x_3 & 0 & y_1 - y_3 & 0 \\ x_1 - x_4 & 0 & y_1 - y_3 & 0 \end{bmatrix} \begin{bmatrix} x \\ 0 \\ y \\ 0 \end{bmatrix} \\ = \begin{bmatrix} d_2^2 - d_1^2 + x_1^2 + y_1^2 - x_2^2 - y_2^2 \\ d_3^2 - d_1^2 + x_1^2 + y_1^2 - x_3^2 - y_3^2 \\ d_4^2 - d_1^2 + x_1^2 + y_1^2 - x_4^2 - y_4^2 \end{bmatrix}. \end{aligned} \tag{8}$$

Therefore, the system observation equation is given as follows:

$$Z_k = CX_k + v_k \tag{9}$$

$$Z_k = \begin{bmatrix} d_{2,k}^2 - d_{1,k}^2 + x_1^2 + y_1^2 - x_2^2 - y_2^2 \\ d_{3,k}^2 - d_{1,k}^2 + x_1^2 + y_1^2 - x_3^2 - y_3^2 \\ d_{4,k}^2 - d_{1,k}^2 + x_1^2 + y_1^2 - x_4^2 - y_4^2 \end{bmatrix}$$

$$C = 2 \cdot \begin{bmatrix} x_1 - x_2 & 0 & y_1 - y_2 & 0 \\ x_1 - x_3 & 0 & y_1 - y_3 & 0 \\ x_1 - x_4 & 0 & y_1 - y_3 & 0 \end{bmatrix} \tag{10}$$

where Z_k is the observation vector, C is the measurement matrix, and v_k is the measurement noise, usually zero-mean, irrelevant white noise.

The algorithm update process is given as follows.

Predicted (A Prior) State Estimate:

$$\hat{x}_{k|k-1} = A\hat{x}_{k-1|k-1}. \tag{11}$$

Predicted (A Priori) Estimate Covariance:

$$P_{k|k-1} = AP_{k-1|k-1}A^T + Q. \tag{12}$$

Optimal Kalman Gain:

$$K_k = AP_{k-1|k-1}A^T + Q. \tag{13}$$

Updated (A Posteriori) State Estimate:

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k\left(Z_k - C\hat{x}_{k-1|k-1}\right). \tag{14}$$

Updated (A Posteriori) Estimate Covariance:

$$P_{k|k} = (I - K_kC)P_{k|k-1}. \tag{15}$$

B. Robust Kalman Filter

When the tag is in the LOS environment, the SKF algorithm can reduce the LOS error very well. However, when a tag is in the NLOS environment, the observations have a large deviation, which will significantly affect the NLOS positioning effect. To solve these problems, some scholars use a robust Kalman filter (RKF) [36]. When the difference between the observations and the predicted value exceeds the set threshold, it is judged to be in the NLOS scene, and the error observation is discarded by modifying the filter gain K as follows:

$$e_k = Z_k - A\hat{x}_{k|k-1} \tag{16}$$

$$K_k^i = \begin{cases} 0, & |e_k^i| > \text{thresh} \\ K_k^i, & |e_k^i| \le \text{thresh}, \end{cases} \quad i = 1, 2, 3 \tag{17}$$

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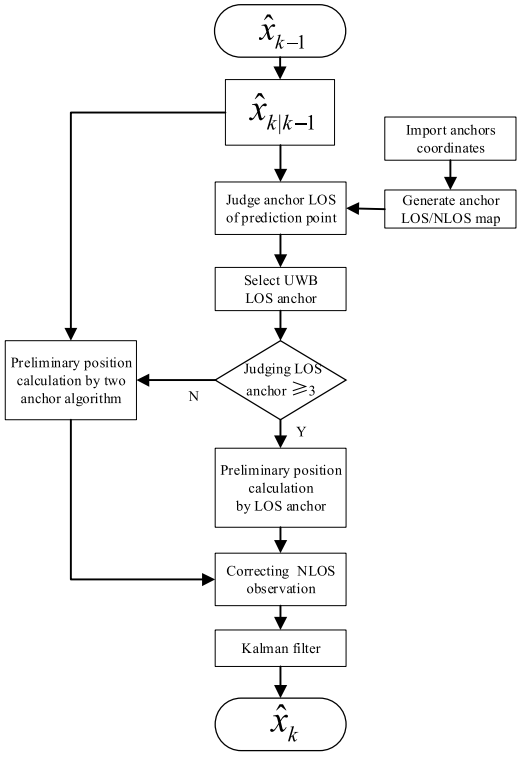


Fig. 5. Algorithm flowchart.

where e_k is the new information vector, e_k^i is the ith value of e_k, K_k^i is the ith column vector of K_k, and thresh is the robust factor.

Through literature review [25], it is difficult to select robust factor for RKF algorithm. Setting the robust factor too large will lose the robust effect, and setting too small will easily cause positioning divergence due to error disturbance. Besides, it is found in the experiment that the RKF algorithm is not effective when the moving target is moving in the serious NLOS area. It will be shown in the following experiment.

C. Map-Modified Kalman Filter

In the SKF algorithm, the NLOS observations affect its positioning in NLOS areas, while in the RKF algorithm, discarding the NLOS observations will affect the results in the case of insufficient observations. Therefore, correcting the NLOS observations can effectively reduce the error. Based on this idea, a map-modified Kalman filter (MKF) algorithm is proposed. When an NLOS anchor is detected at the predicted location, the current observation positioning is preliminarily calculated using other anchors. Then, the observation position is used to correct the corresponding NLOS ranging value, and the SKF algorithm is carried out. The algorithm flowchart is shown in Fig. 5.

It is worth noting that when the number of LOS anchors is only 2, the least squares method is no longer used to solve the positioning, and the position-constrained two anchors auxiliary positioning algorithm presented in this article is used to solve the positioning.

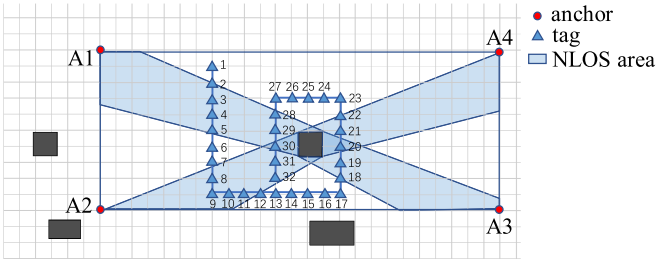


Fig. 6. Calibration position.

TABLE I
GAUSSIAN ERROR OF LOS (m)

Anchor	mean	std
A1	0.062279834	0.042720375
A2	0.133829745	0.117832323
A3	0.096672384	0.049529258
A4	0.118850040	0.073074085

The observations are corrected as the following equation:

$$d_{\text{nlos},k} = \sqrt{\left(\hat{x} - x_{\text{nlos}}\right)^2 + \left(\hat{y} - y_{\text{nlos}}\right)^2} \tag{18}$$

where $d_{\text{nlos},k}$ is the corrected ranging value of the NLOS anchor at time k, $(\hat{x}, \hat{y})$ is the preliminary solution coordinate of the LOS anchor, and $(x_{\text{nlos}}, y_{\text{nlos}})$ is the coordinate of the current NLOS anchor.

IV. Model Simulation

A. Error Calibration

In order to better fit the actual situation, the LOS error (system error) needs to be calibrated by distance measurement. The experiment was performed on the right side of room 309, where four anchors were fixed and tags moved on the corresponding grid points in the order shown in Fig. 6.

The grid interval is 0.5 m, and 750 measurements are made at each location, totaling 32 points [37]. The ranging Gauss error for each anchor is calculated for each point. Notice that some of the ranging values are obtained in the NLOS environment, so after discarding the NLOS error data, calculate the mean value of the LOS error of each anchor, as shown in Table I.

As can be seen from Table I, the anchor ranging error is about 10 cm in the LOS environment, but it is found in experiments that under the NLOS environment, the mean of ranging error is large, reaching meter level, and there is no obvious rule. Because the algorithm used in this article does not use the original NLOS ranging values, the NLOS error is not studied too much in the simulation. The NLOS error is assumed to be ten times the LOS error of the corresponding anchor [37].

B. Simulation Experiment

First, simulate the L-shaped route with single anchor NLOS error. We assume that the speed of tag uniform motion is 0.2 m/s, the sampling frequency is 50 Hz, the walking route is shown in Fig. 7, and the LOS error is added to the simulated

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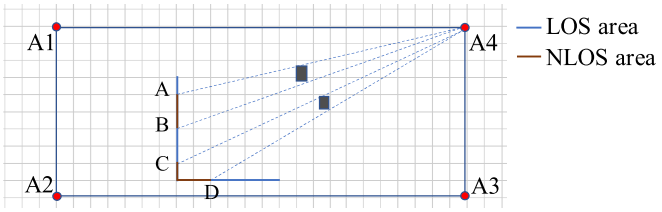


Fig. 7. Simulation experiment 1.

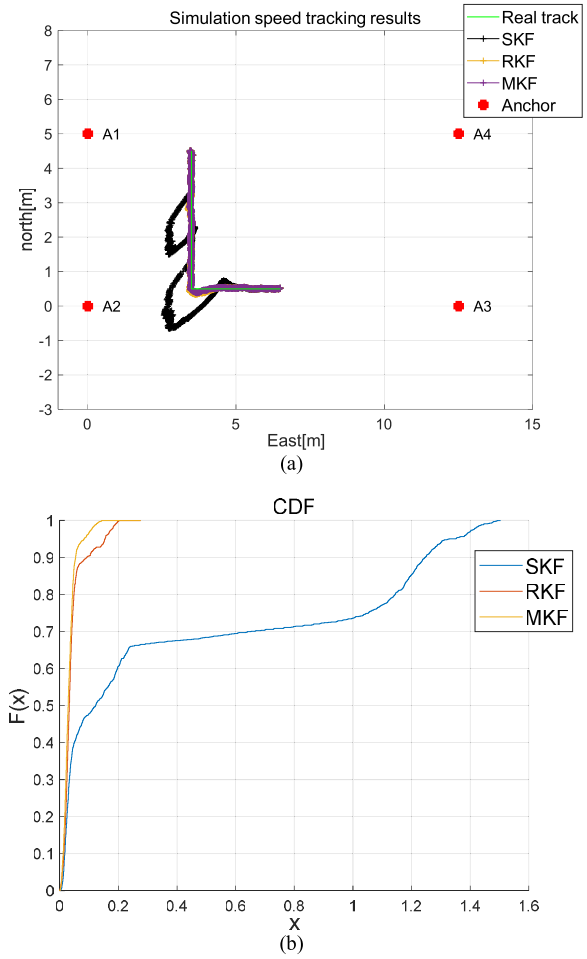


Fig. 8. Simulation experiment 1 positioning results: (a) trajectory and (b) CDF comparison.

ranging values. Among them, A4 anchor is in NLOS area at part of the route, the NLOS segment area is straight line AB segment, and the corner CD segment is shown in Fig. 7. Among them, A4 anchor is in NLOS state at part of the route, the NLOS segment area is straight line AB segment, and the corner CD segment is shown in Fig. 7. Based on adding LOS error and then adding A4 NLOS error to the NLOS area, the simulation is completed.

The following is the positioning solution, using SKF, RKF, and MKF proposed in this article.

Fig. 8 shows the trajectory results and error cumulative distribution functions (CDFs) of the three algorithms, and Table II shows the error statistics of the three algorithms. From the positioning results, it can be seen that the SKF algorithm has an obvious positioning error in the NLOS area due to

TABLE II
COMPARISON OF POSITIONING ERRORS (m)

Algorithm	Mean	Std	Max
SKF positioning error	0.4079	0.4907	1.4593
RKF positioning error	0.0442	0.0413	0.2753
MKF positioning error	0.0330	0.0287	0.2753

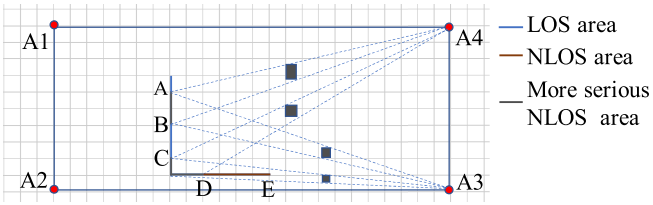


Fig. 9. Simulation experiment 2.

TABLE III
COMPARISON OF POSITIONING ERRORS (m)

Algorithm	Mean	Std	Max
SKF positioning error	0.6675	0.4739	1.6336
RKF positioning error	0.4506	0.4616	1.4894
MKF positioning error	0.0278	0.0232	0.2753

the introduction of NLOS observation values. The positioning error of SKF algorithm is less than 0.5 m in 68%, less than 1.25 m in 90%, 1.46 m in maximum, and 0.41 m in average. Through multiple experiments, it is found that the robust factor 7 is chosen to solve the problem in the RKF algorithm. The positioning error of RKF algorithm is less than 0.09 m in 90%, 0.28 m in maximum, and 0.04 m in average. The MKF algorithm is slightly better than the RKF algorithm, 90% of the positioning error is within 0.05 m, the maximum error is about 0.28 m, and the average error is about 0.03 m. To sum up, it can be considered that the SKF algorithm has poor positioning results when there are three LOS anchors in the NLOS area. The RKF and MKF algorithms have good results in the NLOS area with small difference. The reason is that the NLOS area is relatively single, only one NLOS anchor exists in the NLOS area, and two NLOS anchors do not appear at the same time in the whole experiment path.

To reflect the algorithm differences in complex NLOS environments, the situation that there are two anchors on the route that are in the NLOS area at the same time is simulated, as shown in Fig. 9. Based on single NLOS anchor simulation, A3 anchor is also in the NLOS area, which is the same as A4 anchor. AB and CD segments are two anchors NLOS area, and the DE segment is the A3 anchor NLOS area.

The positioning results and error CDFs of the three algorithms are shown in Fig. 10, and Table III shows the error statistics of the three algorithms.

The experimental results show that the positioning error of SFK in the NLOS area is more obvious, only 35% of the positioning error is within 0.5 m and 90% of the positioning error is within 1.22 m, the maximum error is about 1.63 m, and the average error is 0.67 m. Through multiple experiments, it is found that the RKF algorithm will diverge when using smaller robust factor (less than 10) and lose the robust effect when

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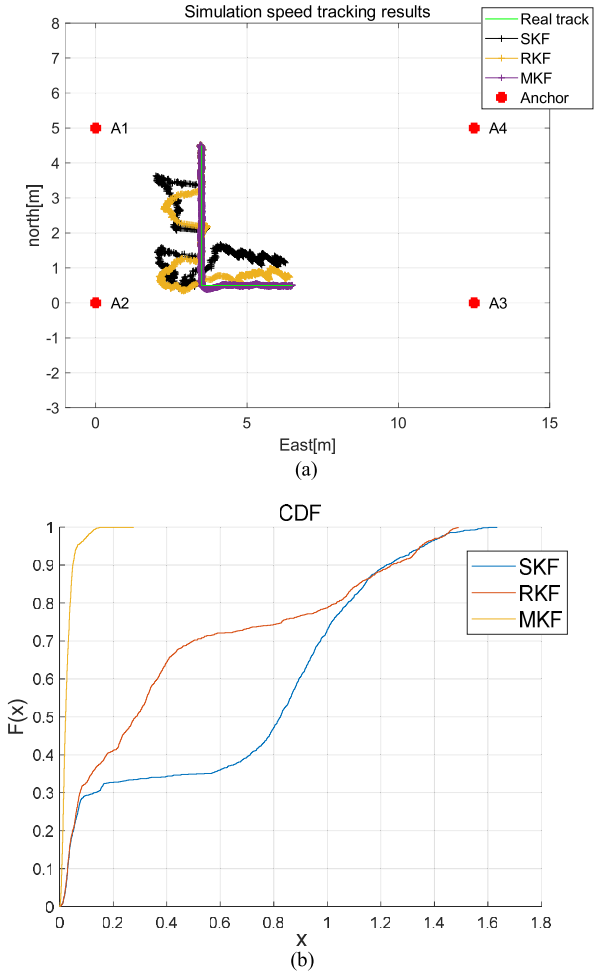


Fig. 10. Simulation experiment 2 positioning results: (a) trajectory and (b) CDF comparison.

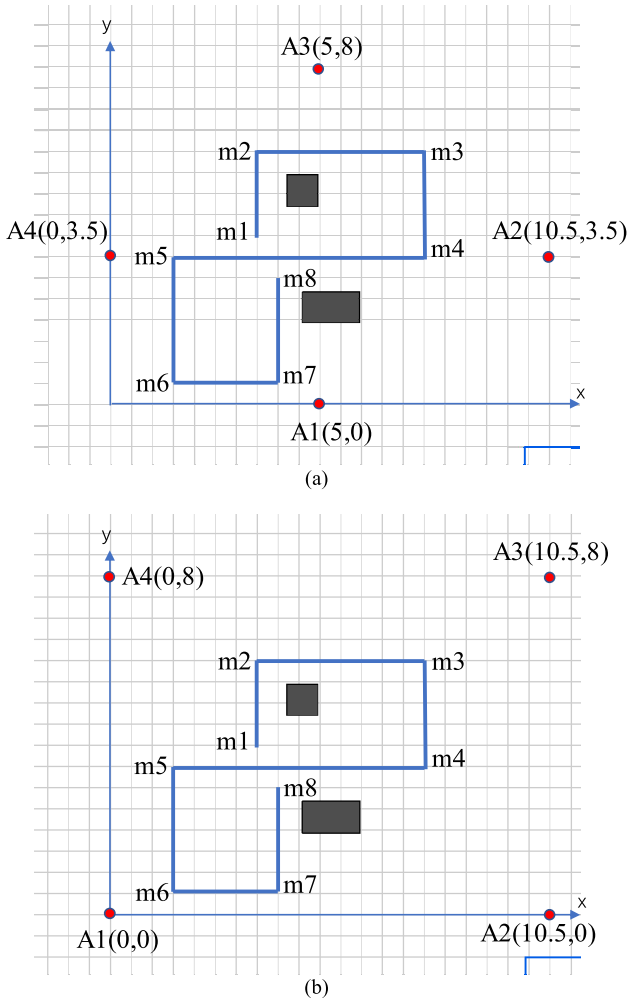


Fig. 11. Tag road map and anchors' placements: (a) scenario 1 and (b) scenario 2.

using larger robust factor (more than 50). In the adjustment process, the robust factor 25 is chosen to solve the problem. The positioning error of RKF algorithm is less than 0.5 m in 70%, less than 1.24 m in 90%, 1.49 m in maximum, and 0.45 m in average. The MKF algorithm is ideal, 90% of the positioning error is within 0.05 m, the maximum error is about 0.28 m, and the average error is about 0.03 m. To sum up, it can be seen that when there are only two LOS anchors in the NLOS area, the SKF and RKF algorithms have poor positioning results due to the lack of valid observation values in the NLOS area, while the MKF algorithm has better positioning results and positioning continuity than the SKF algorithm and RKF algorithm because the MKF algorithm uses the position-constrained two anchors auxiliary positioning algorithm in the NLOS area.

Therefore, the results of simulation positioning experiments on tags moving in LOS/NLOS mixed scenes show that when the environment is open, the solution of SKF algorithm can be very close to the real path, but the positioning accuracy decreases significantly when there are NLOS areas. When there are three or more LOS anchors, the positioning effect of RKF algorithm with appropriate robust factor and MKF algorithm is good. When there are only two LOS anchors, the RKF algorithm positioning effect decreases significantly, and the predicted results cannot be obtained by adjusting the robust factor, while the MKF algorithm effect is still ideal, highlighting the advantages of MKF algorithm in fewer LOS anchors, and verifying the effectiveness of this algorithm.

V. Experiment Verification

A. Experimental Program

It can be seen from the simulation that the positioning effect of the MFK algorithm is better than that of other algorithms in the NLOS area of two anchors at the same time. Therefore, in the actual environment, the environment with more complex spatial structure was selected, and two columns are used as obstacles to shield the environment, of which square column 1 is a solid column and column 2 is a hollow column. In order to demonstrate the generalizability of the algorithm, two experimental scenarios were used with different anchors' placements. The anchors' placements and tag movement in a spatial environment are shown in Fig. 11.

The UWB chips used in the experiment are DW1000 chips, and the host computer uses the NAssistant software of Nooploop Technology Company Ltd., which can directly

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(a) (b) (c)

Fig. 12. Experimental scene data collection: (a) UWB hardware device, (b) experimental collection scene, and (c) host computer data collection.

TABLE IV
COORDINATES OF ANCHORS

scenario 1	Anchor	A1	A2	A3	A4
	Coordinate(m)	(5,0)	(10.5,3.5)	(5,8)	(0,3.5)
scenario 2	Anchor	A1	A2	A3	A4
	Coordinate(m)	(0,0)	(10.5,0)	(10.5,8)	(0,8)

TABLE V
COORDINATES OF COLUMNS

Wall1 corner	Coordinate(m)	Wall2 corner	Coordinate(m)
Lower left corner	(4.32,4.72)	Lower left corner	(4.72,1.97)
Lower right corner	(5.02,4.72)	Lower right corner	(6.05,1.98)
Upper right corner	(5.05,5.45)	Upper right corner	(6.06,2.7)
Upper left corner	(4.32,5.46)	Upper left corner	(4.72,2.68)

output the distance between anchor and tag. The experimental data collection scene is shown in Fig. 12.

The anchor coordinates of the two experimental scenarios are shown in Table IV, and the column coordinates are shown in Table V.

After determining the location coordinates of the environmental obstacle and anchors, the anchor LOS/NLOS map in this environment can be obtained, as shown in Fig. 13, which can be used by the MKF algorithm to judge the LOS anchors. From the anchor LOS/NLOS maps, the tag motion path in two experiments will pass through multiple NLOS areas, including the NLOS areas caused by two NLOS anchors, which makes the scene more complex.

B. Experimental Results and Analysis

Based on the experimental scenario above, the ranging values of four anchors are obtained through UWB motion tags. First, the LOS error table calibrated before is used to eliminate the system LOS error. Then, three positioning algorithms, the SKF described in the literature [17], the RKF described in the literature [36], and the MKF proposed in this article, are used to solve the positioning problem, and the results are analyzed.

First, analyze experimental scenario 1. A sampling frequency of 50 Hz was used in scenario 1, and 2814 samples were obtained in the experimental scene. Fig. 14 shows the results of the tag movement trajectory in the experimental scene, where the green path is the real path, the black path is the SKF algorithm solution track, the yellow path is the RKF algorithm solution track, the purple path is the MKF algorithm solution track, and the red triangles are the anchors' location. For better comparison and analysis, the anchor LOS/NLOS map of Fig. 13(a) and the three localization trajectories are

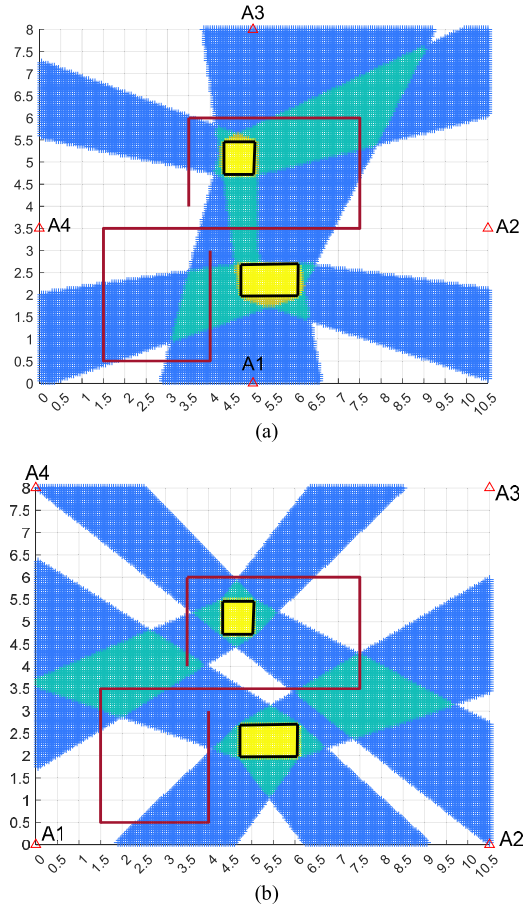


Fig. 13. Anchor LOS/NLOS map: (a) scenario 1 and (b) scenario 2.

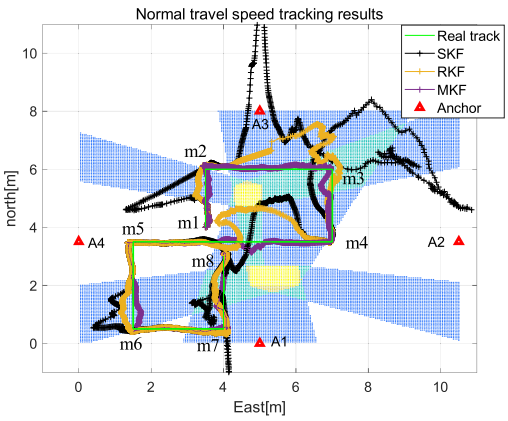


Fig. 14. Trajectory combined with the anchor LOS/NLOS map analysis in scenario 1.

combined into Fig. 14, and the corners of the road segments are marked.

From Fig. 14, it can be seen intuitively that the SKF algorithm solution is the worst and the RKF algorithm is better than SKF, but the solution of two NLOS anchors in the NLOS area at the same time is still not well and the MKF algorithm is the best, basically close to the real track. This is consistent with the predictions of simulation experiments, which shows the effectiveness of the designed algorithm in this article.

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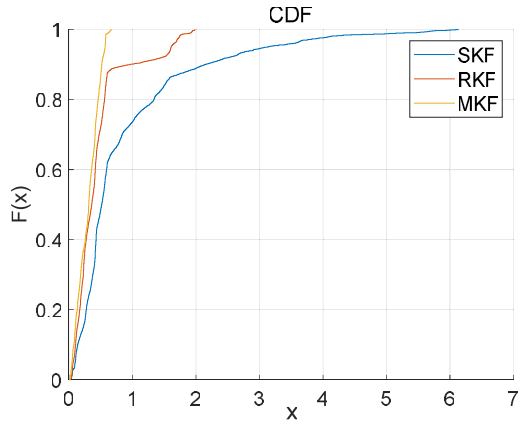


Fig. 15. CDF comparison.

SKF algorithm is a direct solution using the ranging values of four anchors. Because there are NLOS observations in the NLOS area, serious positioning errors will occur. From Fig. 14, it can be seen that the SKF algorithm has a large positioning error in the NLOS area and only maintains good positioning results in the LOS area near the m_5 segment.

The RKF algorithm uses innovation to construct a robust factor matrix, which discards observations that exceed the threshold, so it can optimize the ranging error of NLOS to a certain extent and improve the positioning accuracy. Besides, the selection of threshold needs to be tried repeatedly. If the threshold is too small, the result will diverge, and if the threshold is too large, the robust effect will be lost. In this experiment, several attempts have been made, and the final threshold of 10 is relatively well. From Fig. 14, it can be seen that RKF algorithm is superior to the SKF algorithm in NLOS area, and the error is corrected. However, in the NLOS area of two NLOS anchors, such as near m_3 segment and in the middle of m_4–m_5 path, the optimization effect is not ideal. Positioning errors in the NLOS area of the two anchors in m_7–m_8 are better than those in the front, presumably because the rectangular column on the right is hollow, which makes the NLOS influence of the A2 anchor less significant at the m_7–m_8 path, so no large errors are shown. The reason why RKF algorithm is superior to SKF algorithm is that NLOS ranging values are discarded, but when the effective ranging values are less than 3, the observation values in the Kalman filter algorithm are not enough to correct the prediction values, resulting in prediction errors. The larger the common NLOS area of two anchors, the more positioning errors. Therefore, the positioning accuracy will be reduced due to the existence of NLOS error in the solution process.

The MKF algorithm designed in this article uses the anchor LOS/NLOS map to select the LOS anchors, preliminarily solves coordinates of observation positioning, then uses the observation position to correct the NLOS observations, and then returns to the KF algorithm for the track solution. Therefore, it can be seen from Fig. 14 that in areas where there is only one NLOS anchor, better accuracy can be obtained by using the ranging values of the other three LOS anchors than SKF and RKF. In the area with two NLOS anchors, the position-constrained two anchors auxiliary positioning algorithm proposed in this article can still be used to calculate the observation position, and two NLOS observations can be corrected. Therefore, the positioning solution effect in the area with two NLOS anchors, near m_3 and in the middle of m_4–m_5 path, is obviously better than that of SKF and RKF.

TABLE VI
COMPARISON OF POSITIONING ERRORS (2814 EPOCHS)

Algorithm	Mean(m)	Std(m)	Max(m)	Run time (seconds)
SKF positioning error	0.8859	1.0043	6.1364	0.022326
RKF positioning error	0.4522	0.4212	1.9843	0.027360
MKF positioning error	0.3020	0.1638	0.6721	0.274157

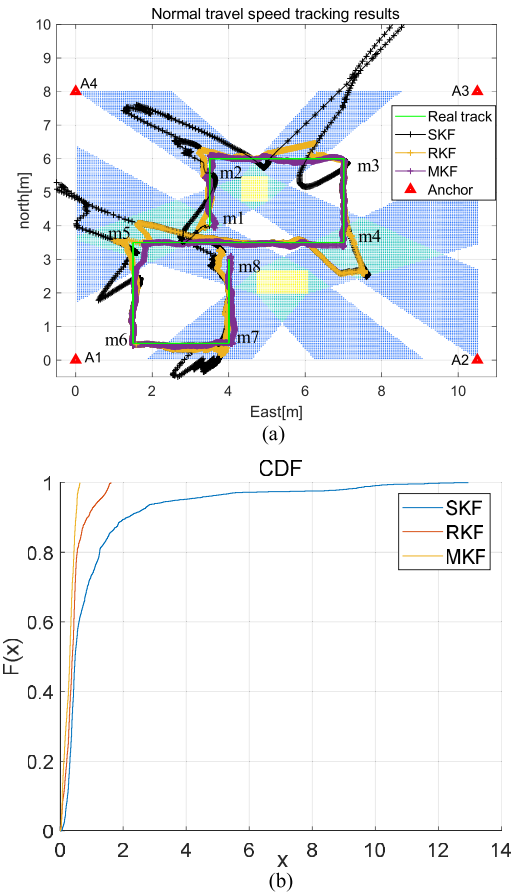


Fig. 16. Scenario 2: (a) trajectory combined with the anchor LOS/NLOS map analysis and (b) CDF comparison.

It can also be seen from the experimental results of scenario 1 in Fig. 15 and Table VI that in this complex NLOS environment, only 74% of the positioning errors of SKF algorithm are within 1 m, the maximum error is about 6.14 m, and the average error reaches 0.89 m. The RKF algorithm has a positioning error of less than 1 m in 90% and a maximum error of about 1.98 m, and the average error is about 0.45 m. The positioning errors of MKF algorithm are all within 1 m, about 90% of them are within 0.50 m, and the average error is about 0.30 m.

The analysis of experimental scenario 2 is similar to that of experimental scenario 1. The trajectory analysis results of the solution are shown in Fig. 16, which is similar to experimental scenario 1. From Fig. 16(a), it can be seen that the SKF algorithm has a large positioning error in the NLOS area and

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TABLE VII
COMPARISON OF POSITIONING ERRORS (2855 EPOCHS)

Algorithm	Mean(m)	Std(m)	Max(m)	Run time (seconds)
SKF positioning error	1.0609	1.6972	12.9544	0.021384
RKF positioning error	0.4411	0.3255	1.6206	0.028594
MKF positioning error	0.2897	0.1559	0.6249	0.284713

only maintains good positioning results in the LOS area near the m_6 segment. The RKF algorithm performs relatively well when the threshold is set to 20. However, in the NLOS area of two NLOS anchors, such as near m_4 segment and m_5 segment, the optimization effect is not ideal. In contrast, by using the MKF algorithm, the positioning solution effect in the area with two NLOS anchors, near m_4 segment and m_5 segment, is obviously better than that of SKF and RKF.

It can also be seen from the experimental results of scenario 2 in Fig. 16(b) and Table VII that in this complex NLOS environment, only 73% of the positioning errors of SKF algorithm are within 1 m, the maximum error is about 12.95 m, and the average error reaches 1.06 m. The RKF algorithm has a positioning error of less than 1 m in 92% and a maximum error of about 1.62 m, and the average error is about 0.44 m. The positioning errors of MKF algorithm are all within 1 m, 90% of them are within 0.50 m, and the average error is about 0.29 m.

From the above two experimental scenarios, it can be concluded as follows.

1) The SKF algorithm directly performs positioning calculations based on all the observation values, achieving high positioning accuracy in LOS conditions. However, in NLOS conditions, the presence of NLOS observation values interferes with the positioning solution and significantly reduces the positioning accuracy.
2) The RKF algorithm, by setting robust factor, distinguishes the NLOS observations to a certain extent and discard them in the positioning solution. Compared with the SKF algorithm, it can reduce the interference of the NLOS observations on the positioning results. However, setting the robust factor in RKF algorithm is difficult, as setting it too large will result in the loss of the robustness effect, while setting it too small will lead to the divergence of the positioning results due to the perturbation of LOS errors. At the same time, the RKF algorithm reduces the number of observations, which may result in significant bias in positioning results when the moving target is in a severe NLOS area, such as when only two base stations are in LOS condition.
3) The MKF algorithm utilizes prior environmental information to obtain the anchor LOS/NLOS map, which can effectively distinguish the LOS and NLOS anchors at the current location, avoiding the problem of threshold factor selection in the RKF algorithm, and the algorithm utilizes the preliminary position coordinates calculated by the LOS anchor to correct the NLOS observation values, avoiding the discarding of the observed values. In addition, the algorithm uses the position-constrained two anchors auxiliary positioning algorithm to select the correct positioning coordinate in the presence of only two LOS anchors so that the validity of observation values in the NLOS region is close to that in the LOS region, ensuring high positioning accuracy even in areas with severe NLOS effects. The overall positioning accuracy of the MKF algorithm is 69.30% higher than that of the SKF algorithm and 33.77% higher than that of the RKF algorithm. The positioning results are ideal, and the positioning continuity is better, which is in line with the expected results of the simulation experiments. Furthermore, the algorithm's generalizability is demonstrated by two different experimental scenarios.

In addition, it is found that the LOS map constraints algorithm and the two anchors positioning algorithm increase the program operation time, for example, reaching 0.27 s in experimental scenario 1, which is ten times longer than the traditional filter algorithm. However, the single calculation time is about 9.74×10^{-5} s, which is still far less than 0.02 s, that is, the time needed for a single sample point in the experiment. For moving target with low real-time requirements, such as pedestrian monitoring and robot monitoring, optimization can be achieved. Moreover, the comparison of the results from two experimental groups indicates that the anchors' placements can also have a certain impact on the positioning results in complex indoor environments.

VI. CONCLUSION

With the development of indoor UWB positioning technology, accurate positioning in indoor NLOS environment has been a research difficulty. This article proposes a novel indoor UWB NLOS correction positioning method based on the anchor LOS/NLOS map. First, the anchor LOS/NLOS map is generated using the prior environmental information, and then, the map is used to select the LOS anchor at the current location, which avoids the difficulty of setting robust factor in the robust algorithm. Second, the preliminary position coordinates calculated by the LOS anchors are used to correct the NLOS observation values, which solves the problem of insufficient observation values in NLOS areas in the robust algorithm. Besides, the method uses the position-constrained two anchors auxiliary positioning algorithm. By using the positioning constraints, the correct positioning coordinates can be selected, which further improves the effectiveness of MKF algorithm in NLOS environments. In this article, the trajectory solution effects of the three filtering algorithms are compared and analyzed through simulation and actual experiments. The results show that the average positioning accuracy of MKF algorithm is 69.30% higher than SKF algorithm and 33.77% higher than RKF algorithm in a complex indoor NLOS environment. The MKF algorithm can effectively reduce the influence of NLOS error of spatial structure on positioning results. It is suitable for high-precision continuous positioning of indoor LOS/NLOS mixed scene.

Currently, the MKF algorithm only considers 2-D planar positioning, which is suitable for pedestrians and robots in fixed floors. In the future work, it can consider 3-D planar positioning to make it suitable for unmanned aerial vehicle in

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indoor environment. The positioning constraint judgment of two anchors in the current algorithm requires that the moving target is on the same side of the line between two LOS anchors. When the moving target crosses the line between LOS anchors, it cannot accurately select the correct coordinates. In the future work, coordinate selection criteria can be further added, such as IMU motion direction information. The MKF algorithm is effective in dealing with NLOS scenarios where there are two or more LOS anchors. In more severe NLOS environments, such as those with only one LOS anchor or no LOS anchor, the algorithm can also be combined with IMU for auxiliary positioning calculation. Besides, the algorithm proposed in this article takes a relatively long time to calculate. It can further optimize the algorithm or improve the computing power so that it can be applied to more real-time moving targets. This article only investigates positioning algorithms under the condition of NLOS caused by static obstacles, without considering the impact of dynamic obstacles. The subsequent research will focus on optimizing the NLOS error caused by dynamic obstacles. Furthermore, the forthcoming research will also explore the impact of anchors' placement on localization accuracy in complex indoor environments.

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